



Security Architecture Project

PROJECT BRIEF

Approvals: This document requires the following approvals. Signed approval forms are filed in the Quality directory in the DataTrust Project Office.

Name	Signature	Title	Date approved	Version
Terrence Stewart	<i>Terrence Stewart</i>	Network Administrator	19/08/2022	1.0

Document control

Document identifier: docx

Document author: Marco Zhou

Version: 1.0

Version date: 19/08/2022

File location: D:\Advanced Diploma\Cyber Security Project\

Revision history

Version	Revision date	Summary of changes
0.1	19/08/2022	Document creation



Document purpose

This document will provide a full and firm foundation for the initiation of the DataTrust Security Architecture Project. The Project Brief is the basis of the Security Architecture Project Plan. Any significant change to the material contained in the project brief will need to be referred to corporate or program management.



Contents

- 1. Background..... 4
- 2. Project definition..... 4
 - 2.1 Project objectives4
 - 2.2 Project scope4
 - 2.3 Project deliverables and outcomes5
 - 2.4 Dependencies and interfaces5
 - 2.5 Assumptions and constraints5
 - 2.6 Outline business case6
 - 2.7 Project approach6
 - 2.8 Project variances6
 - 2.9 Project quality6
 - 2.10 Risk management7
 - 2.11 Project management team structure.....8
 - 2.12 Implementation plan.....8
- Appendix A – Name 9

1. Background

DataTrust has been aware of the prevalence and rise of cyber security issues and incidents in the industry such as attacks, data loss, or system failures. In response to this problem, DataTrust is initiating a cyber-security architecture project that involves the development of a security architecture with the CIA triad applied and implemented. The project can then be applied as a model for client infrastructures or as an industry guideline for security architecture best practices.

2. Project definition

2.1 Project objectives

The main objective of the DataTrust Security Architecture Project is to develop a security architecture that considers the three components of the CIA triad model including confidentiality, integrity and availability. One of the objectives of the project is to develop a high level of security within the security architecture including the protection of data integrity. Another objective of the project is to maintain high availability within the security architecture design to ensure that it is always accessible for users.

2.2 Project scope

2.2.1 Inclusions

- DMZ with Web Server, FTP server and Mail Server
- Firewalls and IDS/IPS
- Antiviruses and security software
- High availability and failover clustering
- Data integrity including protection
- Network logging, monitoring and SIEM
- Identity and Access Management
- Maintenance and patching
- Honeypot
- Staff on boarding and training
- Multi-layer authentication



2.2.2 Exclusions

- Wireless and remote connectivity
- Creating and designing the network infrastructure
- Obtaining equipment and devices
- Cabling
- Purchasing software and obtaining licensing

2.3 Project deliverables and outcomes

The Security Architecture Project deliverables and outcomes will include:

- Vulnerability mitigation system architecture
- Network traffic filtering system
- Authentication and identity management system
- Privacy policies and procedures
- Responsible usage policies
- Backup and disaster recovery procedures

2.4 Dependencies and interfaces

Dependencies and interfaces of the project:

- ROI: They are the client for DataTrust and this project will be completed to provide a security architecture that can be applied to their network infrastructure

2.5 Assumptions and constraints

Assumptions:

- All software is open-source and purchased software is avoided
- The network infrastructure is prepared and ready to apply the security architecture design
- The network infrastructure will be simulated in a virtual environment

Constraints:

- Project deadlines
- Project budget



2.6 Outline business case

The completion of the project will allow for the overall improvement of cyber security across the industry by providing a strong and stable security architecture that mitigates exploitable vulnerabilities or risks, optimizes identity management and authentication, and offers high availability through accessibility and failover clustering technologies.

2.7 Project approach

The project approach will consist of the use of open-source software along with network security devices in order to develop the security architecture. The security architecture will be created with off the shelf products in a phased structure. The selected approach will maximize the chances of project success as the use of a structured development strategy will ensure that each stage of the project is quality assured. The risks of the select approach involves the increased possibility of exploiting known vulnerabilities for the proprietary products such as software.

2.8 Project variances

Tolerances:

- Time: 11 days
- Cost: 5%
- Other: N/A

2.9 Project quality

2.9.1 Customer's quality expectations

The final product of the Cyber Security Project is expected to provide a high quality and fault-proof security architecture that can be applied to secure any network. The product should be able to protect all sensitive information whilst allowing constant access and availability to requested data, and the data should be untouched and untampered with.

2.9.2 Acceptance criteria

The criteria that will be used to ensure that the final product is acceptable for the clients will consist of high availability of 99.999%, and will detect attacks and malicious content without failure.

2.10 Risk management

Executive Summary:

Scope:

Threat/Vulnerability Source:

Vulnerability Tools:

Risk Matrix:

Existing Control Method:

Control Recommendations:



2.11 Project management team structure

Identify as a minimum who shall be the project sponsor and project manager. If possible, identify any additional project team members.

The project team will consist of:

- Project Manager
- Legal Consultant
- Security Administrator
- Network Engineer
- Tester
- Technical Writer

2.12 Implementation plan

Project Phases	Project Documents	Time Estimation	Human Resources
Planning	Project Brief Risk Assessment Plan Gantt Chart	14 days	Project Team
Design	Project Scope Business Case	14 days	Project Manager Legal Consultant Security Administrator Network Engineer Technical Writer
Implementation	Implementation Plan Communication Plan	60 days	Project Team
Evaluation	Quality Assurance Plan	14 days	Project Manager Security Administrator Network Engineer Tester
Hardening	Inventory Audit	7 days	Project Manager Security Administrator Network Engineer Tester
Documentation	Network Logs and Reports Support Documentation	5 days	Project Manager Technical Writer Legal Consultant

Table 1- Use Figure style



Appendix A – Name

Row head	Row head	Row head	Row head	Row head	Row head	Row head
Option 1	Table text	- Table bullet - Table bullet - Table bullet	Table text	- Table Numb-List - Table Numb-List - Table Numb-List	Table text	Table Text-Sml
Option 2	Table text	- Table bullet - Table bullet	Table text	- Table Numb-List - Table Numb-List	Table text	Table Text-Sml
Option 3	Table text	- Table bullet - Table bullet		- Table Numb-List - Table Numb-List	Table text	Table Text-Sml

Table 2- Use Figure style

